THE TRANSLATED PARTIAL-SUM INEQUALITY

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AI assistance was used in adapting and checking the argument and the exposition.

In the Oberwolfach report, Bayart asks: "Is it the case that $\sigma_N \to 0$ as $N \to \infty$?" Here σ_N is the supremum over $f \in H^\infty(\mathbb{C}_0)$ of the infimum of $\sigma > 0$ for which $\|S_N f\|_{H^\infty(\mathbb{C}_\sigma)} \leq \|f\|_{H^\infty(\mathbb{C}_\sigma)}$.

Bayart's problem appears in Oberwolfach Report 51/2017, https://publications.mfo.de/bitstream/handle/mfo/3615/OWR_2017_51.pdf.

Answered: no. Evidence: Sketch.

No. In fact, $\sigma_N = +\infty$ for every $N \geq 2$.

Proof. Let $k = \lfloor \log_2 N \rfloor$ and $m = 2^k$, so $m \leq N < m^2$. Set $f_N(s) = 1 + m^{-s} - m^{-2s}/16$. Its partial sum is $S_N f_N = 1 + m^{-s}$. At height $\sigma > 0$, put $r = m^{-\sigma} \in (0, 1)$. The two supremum norms are the suprema on $|z| \leq r$ of $P(z) = 1 + z - z^2/16$ and $1 + z$, respectively. The latter is $1 + r$.

It remains to show $|P(re^{it})| < 1 + r$. Write $x = \cos t$. If $x \leq 3/4$, then

$$1 + r - |1 + re^{it}| \geq \frac{r(1 - x)}{1 + r} \geq \frac{r}{8},$$

whereas the quadratic term has modulus $r^2/16 < r/8$. If $x \geq 3/4$, direct expansion gives

$$|P(re^{it})|^2 - |1 + re^{it}|^2 = \frac{r^4}{256} - \frac{r^2}{8}(\cos 2t + rx) \leq \frac{r^4}{256} - \frac{r^2}{64} < 0,$$

since $\cos 2t = 2x^2 - 1 \geq 1/8$. The maximum-modulus principle now gives $\|f_N\|_{H^\infty(\mathbb{C}_\sigma)} < \|S_N f_N\|_{H^\infty(\mathbb{C}_\sigma)}$ for every $\sigma > 0$. The defining set of admissible σ is empty, proving the claim. □

Date: 13 October 2026.